

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We explore techniques to bring back extinct species, diets and whether they work at all, NASA editing images and superluminal electrons.



Become a master of dis-information

Researchers at the University of Cambridge have come up with a web application that they hope will vaccinate the players against the influence of fake news. To do this, the application takes you through the process and motivations behind creating and spreading propaganda. Users get to play as a news editor who pushes out fake news stories, while increasing their credibility. <https://digit.in/fake-newsapp>

WHAT'S NEW

Bowhead whales may as well be jazzhead whales

Researchers from the University of Washington have published the largest collection of bowhead whale songs.

The recordings show a huge variety in the types of vocalisations by the whales. The audio recordings were obtained off the coast of Greenland, and include 184 different songs. This is surprising - considering there are believed to be only 200 whales in existence, since they were almost hunted to extinction in the 1600s. The lead author of the paper, Kate Stafford, said, "If humpback whale song is like classical music, bowheads are jazz. The sound is more freeform. And when we looked through four winters of

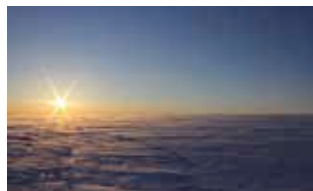


acoustic data, not only were there never any song types repeated between years, but each season had a new set of songs." The researchers cannot yet explain why only the male bowhead whales sing, or why the songs keep changing.

<https://digit.in/bowjazz>

Subglacial lakes could help find life on icy worlds

Scientists from the University of Texas have found the first subglacial lakes in the Canadian high arctic. Previously, subglacial lakes were known to exist in Antarctica and under Greenland. Unlike the other subglacial lakes that are believed to contain freshwater, the subglacial lakes found in the Canadian arctic are believed to be made up of salt water. The scientists believe



that these are the first known saline subglacial lakes, that are isolated from interactions with the outside environment. Studying these lakes could help

identify microbial life elsewhere in the solar system. The lakes found by the researchers are believed to be similar to subglacial lakes found on Europa. The Jovian moons Ganymede and Europa, as well as the Saturnian moons of Titan and Enceladus are the icy worlds that are the most likely locations for microbial life in the solar system.

<https://digit.in/subglakes>



A new source of high energy protons

Nearly 20 years ago, a team of researchers fired a powerful laser at a thin foil and discovered a stream of high energy protons and electrons ejected from the other side. The discovery is now being turned into an experiment that promises to contribute to particle physics. <https://digit.in/heprtns>



Fossils explain ancient butterfly colours

Researchers have studied the tiny ridges and grooves in butterfly fossils from more than 130 million years ago, using an electron microscope. The butterflies are believed to have a metallic brown to golden appearance. <https://digit.in/abfs>



India, US sign agreement to study neutrinos

The US Secretary of Energy Rick Perry, and the Indian Atomic Energy Secretary, Dr. Sekhar Basu signed an agreement on 16 April for collaborations between Fermilab in the US and the India-based Neutrino Observatory. <https://digit.in/inusnu>